

Supporting information. Effects of varying retention tree patterns on ground beetle (Coleoptera: Carabidae) taxonomic and functional diversity. Sean M. Sultaire, Andrew J. Kroll, Jake Verschuyt, Douglas A. Landis, Gary J. Roloff. *Ecosphere*

Appendix S1. Supplemental tables and figures cited in the manuscript

Table S1: Mean climate and elevation (m) for each experimental block, northwest Oregon and southwestern Washington, USA, 2017-2018. Climate variables are PRISM 30-year annual averages with temperature in °C and precipitation in mm. Numbering corresponds to Figure 1.

Block	MeanPrecip	MaxTemp	MnTemp	Elevation
1	1893.19	16.37	10.83	296.83
2	2646.71	15.07	10.1	74.01
3	1939.23	14.2	9.27	606.14
4	1834.91	14.82	9.71	335.79
5	2493.58	13.15	8.37	785.61
6	1797.8	15.01	9.6	431.96
7	2733.67	12.54	8.18	758.57
8	2552.58	13.52	8.35	868.27
9	1989.52	15.71	10.94	159.05
10	3525.35	14.12	9.32	646.13

Table S2: Summary of pitfall trapping effort and carabid beetle captures by retention treatment and year, northwest Oregon and southwestern Washington, USA, 2017-2018. Traps = number of pitfall traps collected (average), Individuals = number of individual carabids (average), Species = number of carabid species (average), Species/Trap Sample = average carabid species per collected trap, and Individuals/Trap Sample = average number of individuals per collected trap.

Treatment ^a	Traps		Individuals		Species		Species/Trap Sample		Individuals/Trap Sample	
	2017	2018	2017	2018	2017	2018	2017	2018	2017	2018
RA	104 (10.4)	141 (14.1)	865 (96)	858 (86)	31 (12)	27 (10)	3.2	2.5	8.3	6.1
UA	108 (10.8)	140 (14.0)	1,030 (103)	1,357 (135)	32 (10)	30 (11)	3.2	2.9	9.5	9.7
S	98 (9.8)	115 (11.5)	650 (65)	836 (92)	30 (10)	30 (11)	3.0	2.7	6.6	7.2
SS	98 (9.8)	138 (13.8)	1,129 (112)	916 (91)	33 (11)	31 (12)	3.5	2.7	11.5	6.6
DS	104 (10.4)	138 (13.8)	1,130 (91)	1,767 (176)	36 (11)	31 (10)	3.5	2.7	10.9	12.8
Total	512 (10.2)	672 (13.4)	4,804 (98)	5,734 (117)	42 (11)	41 (11)	3.3	2.7	9.4	8.5

^aRA= Riparian Aggregated, UA = Upland Aggregated, S = Split, SS = Split with Snags, DS = dispersed with snags

Species	Species Abbr	Body length	Mandible	Disp PCoA	Individuals	Blocks	Stands	Samples
		(mm)	Length/Width					
<i>Agonum muelleri</i>	AGMU	8.28	1.15	-0.41	4	2	3	4
<i>Amara conflata</i>	AMCO	9.95	0.86	-0.40	181	10	24	67
<i>Amara impuncticollis</i>	AMIM	7.79	0.79	-0.41	245	8	18	55
<i>Amara obesa</i>	AMOB	11.57	1.03	-0.06	2	1	1	1
<i>Amara patruelis</i>	AMPA	7.97	0.85	-0.42	68	9	17	27
<i>Amara sanjuanensis</i>	AMSA	7.64	0.94	-0.42	12	5	7	10
<i>Amara spp</i>	AMSP	6.06	0.98	-0.46	5	2	2	2
<i>Anisodactylus binotatus</i>	ANBI	11.65	0.83	-0.36	18	5	7	11
<i>Calathus fuscipes</i>	CAFU	11.87	1.06	-0.05	69	5	8	21
<i>Carabus nemoralis</i>	CANE	22.83	1.29	0.34	10	2	2	4
<i>Carabus taedatus</i>	CATA	20.38	1.19	0.42	162	2	10	84
<i>Cychrus tuberculatus</i>	CATU	20.97	1.42	0.42	243	10	41	171
<i>Harpalus affinis</i>	HAAF	11.02	0.89	-0.38	75	2	4	13
<i>Harpalus animosus</i>	HAAN	12.78	0.84	-0.34	63	4	8	37

<i>Harpalus cordifer</i>	HACO	8.39	0.82	-0.07	168	10	28	94
<i>Harpalus spp</i>	HASP	9.40	0.86	-0.07	467	10	44	245
<i>Lebia viridis</i>	LEVI	4.00	0.95	-0.47	1	1	1	1
<i>Leistus ferruginosus</i>	LEFE	8.96	0.56	-0.36	10	5	8	9
<i>Microlestes nigrinnis</i>	MINI	2.83	0.81	-0.10	1	1	1	1
<i>Nebria brevicollis</i>	NEBR	11.31	0.80	-0.34	16	9	10	12
<i>Notiophilus sylvaticus</i>	NOSY	4.88	0.89	-0.09	19	9	15	18
<i>Omus audouini</i>	OMAU	15.13	1.09	0.27	940	4	10	90
<i>Omus dejeani</i>	OMDE	19.52	1.07	0.35	1050	8	33	383
<i>Platynus ovipennis</i>	PLOV	10.89	1.04	0.18	7	5	6	7
<i>Promecognathus crassus</i>	PRCR	11.77	1.49	0.16	127	9	24	90
<i>Pterostichus adstrictus</i>	PTAD	11.49	1.12	-0.37	6	1	2	5
<i>Pterostichus algidus</i>	PTAM	13.08	1.03	0.18	2296	10	37	421
<i>Pterostichus amethystinus</i>	PTAM	12.93	0.98	0.18	45	6	20	43
<i>Pterostichus crenicollis</i>	PTCR	17.29	1.14	0.27	46	7	9	31
<i>Pterostichus herculaneus</i>	PTHE	14.06	0.99	0.12	666	6	23	272

<i>Pterostichus infernalis</i>	PTIN	9.17	1.08	0.13	384	3	14	147
<i>Pterostichus lama</i>	PTLA	23.30	1.03	0.40	529	10	44	336
<i>Pterostichus nigracaeruleus</i>	PTNI	11.02	1.19	0.13	1	1	1	1
<i>Pterostichus pumilis</i>	PTPU	8.41	1.13	0.13	523	7	34	246
<i>Pterostichus rothi</i>	PTRO	10.16	1.17	0.13	2	2	2	2
<i>Pterostichus spp</i>	PTSP	12.63	1.07	0.15	19	8	13	16
<i>Scaphinotus angulatus</i>	SCAL	17.54	1.82	0.34	9	5	7	9
<i>Scaphinotus angusticollis</i>	SCAN	19.88	1.42	0.46	940	8	20	152
<i>Scaphinotus marginatus</i>	SCMA	12.73	1.67	0.26	124	8	24	78
<i>Syntomus americanus</i>	SYAM	2.91	0.96	-0.10	54	10	20	39
<i>Tanystoma sulcata</i>	TASU	10.67	1.02	0.14	2	1	2	2
<i>Trachypachus holmbergi</i>	TRHO	4.61	0.71	-0.47	1230	10	39	268
<i>Unknown1</i>	NA	3.03	0.98	NA	2	1	1	1
<i>Unknown2</i>	NA	5.64	0.91	-0.46	3	1	1	2
<i>Unknown3</i>	NA	2.77	1.07	-0.50	1	1	1	1

<i>Unknown5</i>	NA	4.85	0.86	-0.50	1	1	1	1
<i>Zacotus mathewsi</i>	ZAMA	15.13	0.99	0.19	367	10	35	186

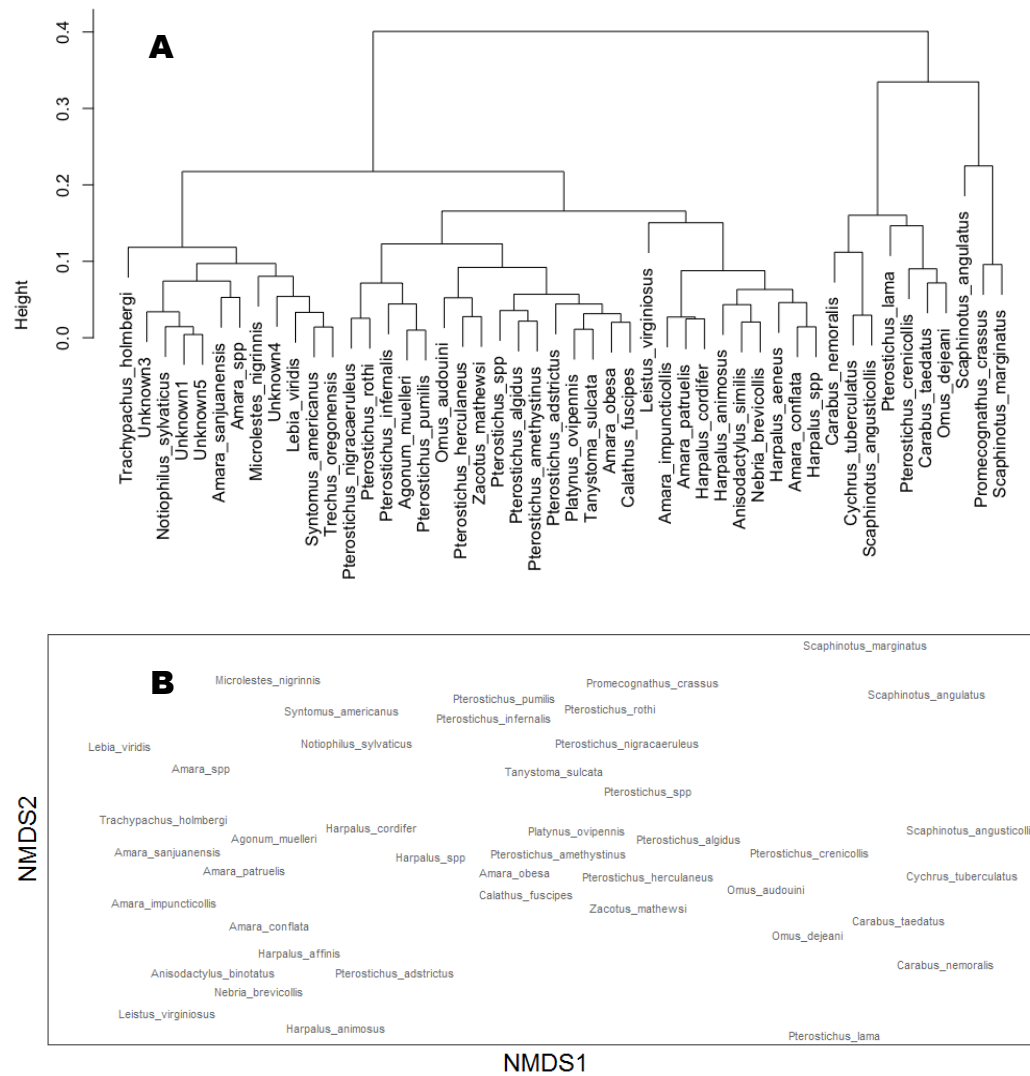


Figure S1: A) Functional dendrogram and B) NMDS ordination for all carabid species and morphospecies sampled in 50 clearcut treatment stands, northwest Oregon and southwestern Washington, USA, 2017-2018. Dendrogram and ordination were constructed from pairwise dissimilarity between each species based on body length, mandible length to width ratio, wing type, and leg length based on Gower's distance.